Polynomial Regression Model Selection

BT2024015 - Assignment Report

1. Problem Understanding and Objective

The assignment requires polynomial regression models for two personalized datasets. The objective is to predict a continuous target for unseen test data while selecting a polynomial degree that balances predictive accuracy and model complexity. Mean Squared Error (MSE) and R-squared (R^2) are used to evaluate validation performance. Degrees 1-10 are considered for var1 and degrees 1-20 for var2.

For var1, the input consists of six operational variables (x1-x6) associated with a geothermal steam-turbine system. For var2, the input consists of three spatial variables (x1-x3) from a 3D geological survey, with the target representing a Thermal Anomaly Score. The second problem is therefore allowed a wider polynomial search because its relationship may require substantially higher-order terms.

2. Model Development Strategy

Four regression approaches are evaluated at every polynomial degree: ordinary Linear Regression, Ridge Regression, Lasso Regression, and Elastic Net Regression. Instead of assuming in advance which degree or regularizer will work best, the code evaluates every permitted degree and lets cross-validation determine the most suitable configuration.

Each model is implemented as a single pipeline. The original input variables are first standardized, then PolynomialFeatures generates polynomial and interaction terms up to the selected degree, and the expanded feature matrix is standardized again before fitting the regression model. The second scaling step is particularly important for Ridge, Lasso, and Elastic Net because regularization penalizes coefficient magnitudes; scaling the generated terms makes that penalty more comparable across features.

Model comparison uses 5-fold K-fold cross-validation with shuffling and random_state=42. The same fold structure is used throughout the search, so CV MSE values are directly comparable across degrees and model types. MSE is the main selection metric because lower MSE directly represents lower average squared prediction error, while CV R^2 is reported as an additional measure of explained variance.

3. Regularization and Hyperparameter Selection

Polynomial expansion can increase the feature space very rapidly. In var1, the number of generated polynomial terms grows from 6 at degree 1 to 8007 at degree 10. In var2, it grows from 3 at degree 1 to 1770 at degree 20. This makes an unregularized high-degree model much more sensitive to correlations among polynomial terms and to small changes in the training folds.

Ridge uses an L2 penalty and searches alpha values from 10^-4 through 10^4 in powers of ten. Lasso uses an L1 penalty and selects alpha through LassoCV, which can drive some coefficients exactly to zero. Elastic Net combines L1 and L2 penalties and searches multiple L1-ratio values using ElasticNetCV. These methods allow the polynomial degree to become expressive without relying entirely on an unconstrained least-squares fit.

4. Complexity-Aware Selection Rule

After all configurations are evaluated, the code first identifies the absolute minimum cross-validation MSE. It then sets a tolerance threshold equal to 102% of that minimum. Every configuration whose CV MSE is within this threshold is considered practically close to the best result, and among those configurations the lowest polynomial degree is selected.

This is a deliberate complexity preference rather than a different evaluation metric. The reason is that a tiny reduction in validation MSE may not justify adding several more polynomial orders and the corresponding increase in feature count. The rule therefore prefers a simpler model when its predictive performance is essentially indistinguishable from the best observed configuration. This makes the final choice more defensible than simply selecting the largest degree or blindly choosing the absolute minimum.

5. Results and Detailed Rationale - var1

var1 contains 1000 training observations, 1000 test observations, and six original features. Degrees 1-10 were evaluated. The main pattern is a strong improvement from low degrees to degree 5, followed by instability for unregularized regression and little useful improvement from higher-degree regularized models.

Model	Best Degree	CV MSE	CV R^2
Linear	4	0.779808	0.92974
Ridge	5	0.480990	0.95666
Lasso	5	0.336393	0.96969
Elastic Net	5	0.337005	0.96963

Underfitting at low degrees. Degree 1 gives a CV MSE of about 9.74 for Lasso/Elastic Net, showing that a simple linear relationship cannot represent the dataset well. Increasing the degree allows nonlinear and interaction effects to be represented. By degree 3 the error falls to roughly 0.91, and by degree 4 it drops further to roughly 0.55. This large reduction indicates that the target depends on nonlinear combinations of the six input variables.

Degree 5 is the useful accuracy region. At degree 5, the regularized models reach their strongest validation performance. Lasso achieves the lowest overall CV MSE of 0.336393 with CV R^2 of 0.96969. Elastic Net is extremely close at 0.337005, but it is still slightly worse, so there is no reason to replace Lasso with the more general mixed penalty in this dataset.

Why higher degree is not better. For ordinary Linear Regression, degree 6 is a major failure point: the CV MSE jumps to 378.019 and CV R^2 becomes -33.061. The later linear degrees remain very poor, with CV MSEs around 8 or higher. This shows that increasing polynomial degree without regularization makes the expanded design matrix numerically unstable and highly sensitive to the folds. Lasso and Elastic Net avoid this catastrophic behavior, but after degree 5 their CV MSE stays around 0.35-0.38, so additional polynomial complexity produces no meaningful validation gain.

Final var1 choice: Degree 5 Lasso. The selected alpha is approximately 0.007127, with 148 non-zero coefficients. The full-training fit gives MSE 0.237191 and R^2 0.97863. The model therefore captures the nonlinear structure while still using L1 regularization to eliminate many generated terms.

6. Why Lasso Is the Best Regularizer for var1

The regularization results are informative beyond the single best score. Ridge improves substantially over ordinary linear regression, but its best CV MSE (0.480990) remains well above Lasso and Elastic Net. This suggests that simply shrinking all coefficients is not enough. Lasso performs better because its L1 penalty can set unnecessary polynomial coefficients exactly to zero, effectively performing feature selection inside the expanded polynomial space. With 148 non-zero coefficients at the selected degree, the final model uses only a subset of the available terms while retaining the best observed validation accuracy.

7. Results and Detailed Rationale - var2

var2 contains 1000 training observations, 1000 test observations, and three original features. Degrees 1-20 were evaluated. In contrast to var1, validation error continues improving into the higher-degree region, which supports the need for a more expressive polynomial representation.

Model	Best Degree	CV MSE	CV R^2
Linear	9	0.266775	0.99467
Ridge	13	0.241197	0.99518
Lasso	19	0.305200	0.99391
Elastic Net	18	0.311575	0.99378

Strong underfitting at low degrees. Degrees 1-4 are clearly insufficient. CV MSE decreases from 37.78 at degree 1 to about 3.48 at degree 4, showing that the relationship is far from linear. The model keeps improving as more polynomial terms are introduced; degree 8 gives MSE about 0.2657 and degree 10 gives about 0.2506.

Ridge gives the best validation curve. The absolute best configuration is degree 13 Ridge with CV MSE 0.241197 and CV R^2 0.99518. This is better than the best ordinary linear model at degree 9 (0.266775), and it is also better than the best Lasso and Elastic Net results. The difference is important because var2 contains a large high-degree feature space, where L2 shrinkage can stabilize many correlated polynomial terms without discarding as many of them as L1 regularization does.

Why ordinary Linear Regression fails at high degree. Although linear regression benefits from increasing degree initially, it becomes extremely unstable later. Its CV MSE rises to 84.838 at degree 14 and eventually reaches 906.768 at degree 20, with CV R^2 becoming strongly negative. This is a clear indication that unconstrained least squares cannot safely handle the high-degree polynomial representation on this dataset.

Why Lasso and Elastic Net are not selected. Both regularized models remain much more stable than ordinary linear regression, but their best errors are still higher than Ridge: 0.305200 for Lasso and 0.311575 for Elastic Net. Therefore, their additional sparsity or mixed regularization does not provide a validation advantage large enough to justify choosing them over Ridge for var2.

8. Why Degree 11 Ridge Was Selected for var2

The absolute minimum CV MSE is achieved by degree 13 Ridge at 0.241197. However, the final selection rule intentionally does not always choose that exact minimum. The 2% tolerance produces a threshold of approximately 0.24602. Degree 12 Ridge has CV MSE 0.242342 and degree 11 Ridge has CV MSE 0.243897; both are inside the threshold.

Among these near-optimal configurations, degree 11 is the lowest degree, so the code selects it. The accuracy sacrifice is very small: compared with degree 13, the CV MSE increases from 0.241197 to only 0.243897, a difference of 0.002700. In return, the polynomial representation is simpler because it contains fewer high-order terms. This is exactly the situation the complexity-aware rule is intended to handle.

Final var2 choice: Degree 11 Ridge with alpha = 1. Its CV MSE is 0.243897 and CV R^2 is 0.99513. When fitted on the complete training data, it achieves MSE 0.16176 and R^2 0.99677.

9. Comparative Interpretation

Problem	Final Model	Degree	CV MSE	Main Rationale
var1	Lasso	5	0.336393	Best validation result with useful sparsity; higher degrees add little benefit
var2	Ridge	11	0.243897	Within 2% of best degree-13 Ridge while using a simpler polynomial

The two datasets therefore require different modelling decisions. For var1, degree 5 is the point where nonlinear flexibility is sufficient and Lasso gives the lowest validation error. For var2, the useful degree is much higher, and Ridge is preferable because the expanded feature space remains stable under L2 regularization. The results demonstrate why the polynomial degree and regularization method should be selected from validation behavior rather than chosen once for both problems.

10. Final Pipeline, Prediction, and Conclusion

Once the final configuration is selected for each problem, the chosen pipeline is fitted again using the complete training dataset. The same fitted preprocessing and regression steps are then applied to the corresponding test features to generate the required predictions. The resulting prediction files are BT2024015_pred_var1.csv and BT2024015_pred_var2.csv.

Final Model Summary

Problem	Selected Model	Degree	Regularization	CV MSE	CV R^2	Train MSE	Train R^2
var1	Lasso	5	alpha = 0.007127	0.336393	0.96969	0.237191	0.97863
var2	Ridge	11	alpha = 1	0.243897	0.99513	0.16176	0.99677

Conclusion

The final solution combines polynomial feature expansion, two-stage standardization, 5-fold cross-validation, regularization, and a 2% complexity-aware selection rule. The experiments show two different forms of behavior. var1 benefits strongly from nonlinear terms up to degree 5, after which additional complexity gives little or unstable improvement; Lasso is therefore the strongest choice. var2 continues to benefit from higher-order terms, but unconstrained linear regression becomes unstable and Ridge provides the best validation performance. The final degree-11 Ridge model is chosen instead of the absolute degree-13 minimum because it stays within 2% of the best CV MSE while being simpler.